

$$\{Q_a, \bar{Q}_b\} = -\sigma^{\mu}_{ab} P_\mu$$

$$\partial_b = \frac{\partial}{\partial \theta^b}$$

$$Q_a = \partial_a, \quad \bar{Q}_b = -\partial_b + i \theta^a \sigma^{\mu}_{ab} P_\mu \quad | \quad P_\mu = -i \partial_\mu$$

$$\{ \partial_a, \partial_b \} f(\theta^2 \bar{\theta}^2)$$

$$\underset{A}{=} (\partial_a \partial_b + \partial_b \partial_a) \theta^c \theta_c \bar{\theta}_c \bar{\theta}^c$$

$$\begin{aligned} \textcircled{A} &= \theta^2 \partial_b \bar{\theta}_c \bar{\theta}^c - \theta^2 \theta_c \delta^c_b \\ &= \theta^2 \partial_b \bar{\theta}_c (\bar{\theta}^c) - \theta^2 \theta_c \theta^2 \\ &= \theta^2 \theta_c \delta^c_b \partial_b \bar{\theta}^c \bar{\theta}^c - \theta^2 \theta_b \\ &= \theta^2 (e^c_b \bar{\theta}^c - \theta_b) \\ &= -2 \theta^2 \bar{\theta}_b \\ &= 4 \theta_a \bar{\theta}_b \end{aligned}$$

$$\begin{aligned} \textcircled{B} \quad \partial_b \partial_a (\theta^2 \bar{\theta}^2) &= -2 \partial_b (\theta_a \bar{\theta}^2) \\ &= (2 \theta_a) (-2 \bar{\theta}_b) \\ &= -4 \theta_a \bar{\theta}_b \end{aligned}$$

$$\therefore \{ \partial_a, \partial_b \} f(\theta^2 \bar{\theta}^2) = 0$$

→ This justifies our exclusion of this term,

$$\{ \partial_a, \theta^b \sigma^{\mu}_{ba} P_\mu \} = -\sigma^{\mu}_{aa} P_\mu \quad (\text{off-shell})$$

→ On-shell high energy limit

$$P_{ba} = -[P]_b \angle p|_a$$

$$\{ \partial_a, \theta^b [P]_b \angle p|_a \} = [P]_a \angle p|_a$$

$$Q \equiv \theta^b [P]_b \quad | \quad \frac{\partial}{\partial \theta^a} = \frac{\partial \mathcal{H}}{\partial \theta^a} \frac{\partial}{\partial \mathcal{H}} = [P]_a \frac{\partial}{\partial \mathcal{H}}$$

Q

$$\{ |P\rangle_a \frac{\partial}{\partial M}, M \langle P |_{\bar{a}} \} = |P\rangle_a \langle P |_{\bar{a}}$$

$$\Leftrightarrow \left\{ \langle P |^a \frac{\partial}{\partial M}, M |P\rangle^{\bar{a}} \right\} = |P\rangle^{\bar{a}} \langle P |^a$$

$$Q^a \equiv \langle P |^a \frac{\partial}{\partial M} \quad | \quad Q^{\bar{a}} \equiv M |P\rangle^{\bar{a}}$$

(On-shell supersymmetry charges for $N=1$)